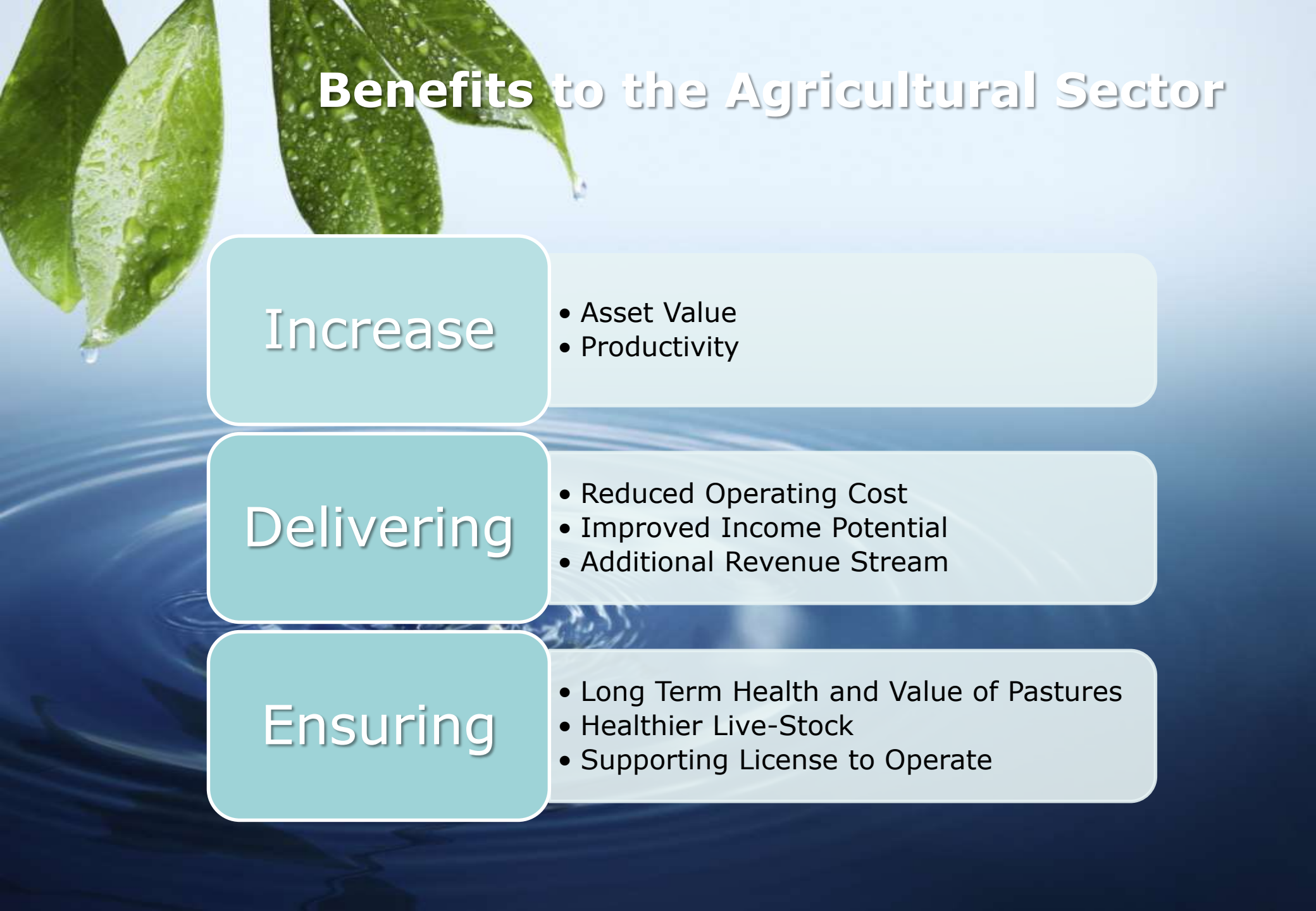


FLOWFORM WATER

ECO-TECHNOLOGY

Enhance Water Quality
Transforming Animal Effluent into Organic Fertilizer

*Farmer's guide contributing to a
profitable and sustainable future*



Benefits to the Agricultural Sector

Increase

- Asset Value
- Productivity

Delivering

- Reduced Operating Cost
- Improved Income Potential
- Additional Revenue Stream

Ensuring

- Long Term Health and Value of Pastures
- Healthier Live-Stock
- Supporting License to Operate

FLOWFORM SUPPORTING FARMERS

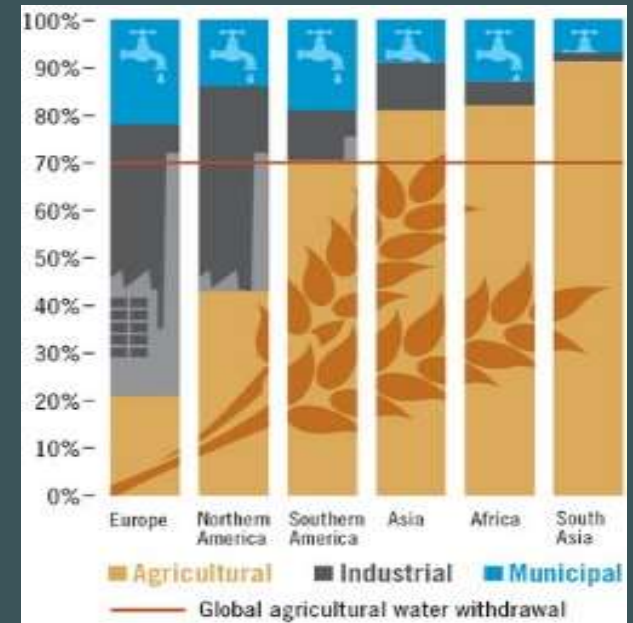
Agricultural Sector is the single biggest water consumer, around 70% of the planet's accessible resource, resulting in deterioration of our eco-system.

Flowform is an eco-friendly, sustainable, low cost water treatment technology to help you re-use water and resolve cattle's effluent treatment challenges.

Flowform mimics a natural process, lowering cost, providing an efficient solution for farmers to develop an effective land and water management strategy, improving your overall farm performance.

Flowform produces an organic plant based fertilizer, replacing many chemical based products, resulting in:

- Enriching your soil, rather than continuing it's depletion
- Highly oxygenated liquid, improves absorption of nutrients by plant roots
- Delivering appropriate pH levels and reducing soil acidification
- Restoring the natural environment of your soil for animal and microbe life
- Reducing disease and sustaining healthy pastures.



THE CHALLENGE

The impact of animal effluent on fresh water worldwide is acknowledged to be a Serious Challenge for a variety of reasons:

Unsustainable practices

Farmers producing organic waste have to dispose of this on their own land or send it to an effluent dump, both of which are costly options. Costly for the farmer's land as the pasture and soil have to deal with transformation of the sludge.

Toxic and pungent Smelling gases are generated

The decomposing activity of microorganisms living in anaerobic organic waste generate pungent smell and toxic gases. The wrong strategy can have costly implications.

Soil acidification

Intensive animal farming, especially dairy farming, has resulted in changed pH balances in soils, resulting in depleting soil quality, reducing efficient use of land ultimately affecting your land value.

THE CHALLENGE continued

Dead Zones in deltas

Nitrification caused by effluent drainage into artesian waters, spreads through rivers to the sea, causing massive die off and loss of oxygen in salt water environments, causing all aerobic life forms to die.

Biochemical oxygen demand levels

Most animal slurries have a high biochemical oxygen demand (BOD) which if entering a watercourse after application to land can deplete the available oxygen content in water and result toxic aquatic environment. High BOD waste when added to wet soils increases anaerobic conditions in the soil depleting oxygen, resulting in deterioration of the surrounding ecosystem, Affecting the potential of your License to Operate

Further reference: European Commission-Directorate-General for Environment :
http://ec.europa.eu/environment/waste/studies/compost/landspreading_4-6.pdf

THE FLOWFORM DIFFERENCE

Flowform® Technology has a unique figure8 flow, in one meter of it's technology it can create water quality improvements that a natural stream takes 10 meters to achieve, **efficiently transforming anaerobic slurry into aerobic liquid fertilizer.**



The presence of aerobic bacteria in the treated slurry to be sprayed on your paddock:

- allows the immediate return of your animals, saving you time and money
- avoids pungent smelly anaerobic gases thus increasing the quality and wellbeing of your livestock
- results in faster vegetation growth due to high dissolved oxygen (DO) levels, increasing your yield value and ensuring healthier livestock

THE SOLUTION

Flowform is a Versatile and Sustainable Low Cost water technology, that **changes poor quality stagnant water into a high quality natural state**, by introducing (1) laminar, (2) harmonic and (3) turbulent water movements.

The science is straightforward. Effluent not constantly flowing will become stagnant with little to zero DO, causing numerous symptomatic problems for farmers.



Typically farmers use costly and ineffective polluting chemicals and unsustainable bubbling machinery, that does not solve the original problem which is water stagnancy and lack of retained oxygen.

Flowform® is the only cost effective technology that deals with the root cause by re-introducing Dynamic Flow, which in nature oxygenates, restructures and energizes water faster and more effectively than any other solution.

YOUR PROBLEM SOLVED IN A NUTSHELL

The Issue:

- Disposing of untreated rotting waste, causing ecologically unbalanced environments, polluting natural land and water ecology, producing pungent smelly effluent full of deadly pathogens



The Cause:

- This is caused by the high level of organic waste in the effluent that in the absence of oxygen turns anaerobic and starts rotting

The Solution:

- Increasing the oxygen level in the rotting effluent, using highly efficient Dynamic Flow. Aerobic decomposing is triggered that stops the pungent smell, kills the deadly pathogens resulting in a useful liquid fertilizer full of nutrients ready to be sprayed on paddocks.



FLOWFORM THE ULTIMATE CHOICE

It turns anaerobic animal effluent waste into **usable aerobic fertilizer** to be sprayed onto your pastures and crops right after processing saving you time and money

Provides quicker **recovery of pastures** with **faster nutrient absorption** by plants, allowing your animals to return faster to paddocks

Removes decomposing **pungent smell** from stagnant animal effluents which reduces disease while creating healthier pastures

Stimulates plant growth as the energized water increases root surface area. This highly oxygenated water maintains and supports healthy soil environment resulting in higher crop yield

Flowform revitalized **drinking water improves the quality and health properties preferred by animals** over tap water

Strengthens organic farming methods, via soil improving compost teas and preparations ultimately increasing the value of your property



FLOWFORM TREATMENT TABLE AND PRICES

Model	Flowrate	Liquid type	Max. size Pond M ³	Days - zero D.O to 3 ppm & no smell	Days - zero DO to 6 ppm	D.O increase whole pond per day	Whole pond turnover rate	Number of FF units
Glonn	80 lpm	Primary effluent	500	20	40	0.15 ppm	Every 4 days	Seven
Vortex	250 lpm	Primary effluent	1,500	20	40	0.15 ppm	Every 4 days	Seven
Glonn	80 lpm	Secondary effluent	500	10	20	0.30 ppm	Every 4 days	Seven
Vortex	250 lpm	Secondary effluent	1,500	10	20	0.30 ppm	Every 4 days	Seven
Greenhouse	7 lpm	Clear water	40	5	10	0.60 ppm	Every 4 days	Seven
Glonn	80 lpm	Clear water	500	5	10	0.60 ppm	Every 4 days	Seven
Vortex	250 lpm	Clear water	1,500	5	10	0.60 ppm	Every 4 days	Seven

Price information:

Greenhouse 7 units: USD 150,00 (excluding pump and VAT)

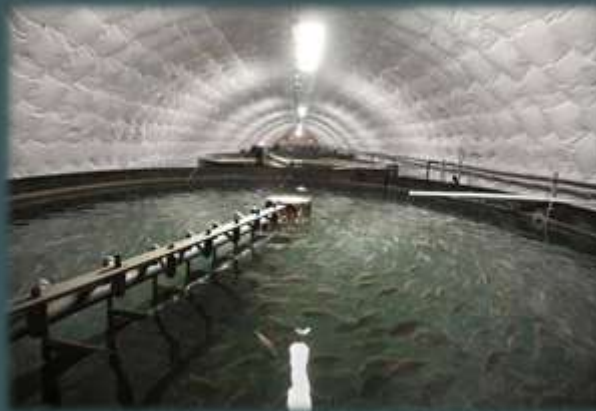
Glonn 7 units: USD 1.800 (including metal frame, excluding pump and VAT)

Vortex 7 units: USD 3.600 (including metal frame, excluding pump and VAT)

ADDITIONAL APPLICATIONS OF FLOWFORM TECHNOLOGY



Compost tea making



Fish farming



Drinking water treatment



Wine making



Energized irrigation

FLOWFORM PRODUCTS

Greenhouse application

- energizing and increasing the oxygen level of irrigation water used in horticultural greenhouses
- stirring biodynamic preparations and soil biology improving agents
- compost tea preparation

Technical data

- Height: 16 cm
- Width: 16 cm
- Length: 25 cm
- Flow rate: 7 liters per minute
- Material: PETG
- Weight: 0,1 kg per piece
- Price: USD150/7 units



Small farm effluent application

- turning animal effluent into liquid fertilizer on smaller farms
- energizing and increasing the oxygen level of irrigation water used in horticultural greenhouses
- stirring biodynamic preparations and soil biology improving agents
- compost tea preparation

Technical data

- Flowrate: 80 lit/min
- Length: 70 cm/Glonn unit
- Width: 50 cm/ Glonn unit
- Height: 15 cm/ Glonn unit
- Weight: 1 kg/ Glonn unit
- Material: PETG
- Price: USD165/single unit



Small farm effluent application

- turning animal effluent into liquid fertilizer on smaller farms
- energizing and increasing the oxygen level of irrigation water used in horticultural greenhouses
- stirring biodynamic preparations and soil biology improving agents
- compost tea preparation

Technical data

- Flowrate: 80 lit/min
- Length: 70 cm/Glonn unit
- Width: 68 cm/ Glonn unit
- Height: 16cm/ Glonn unit
- Weight: 3 kg/ Glonn unit
- Material: fiber glass reinforced food safe plastic
- Price: USD250/single unit



GLONN

Large farm effluent application

- turning animal effluent into liquid fertilizer on high capacity farms (dairy sheds, poultry, piggeries, abbatoirs)
- energizing the irrigation water in agriculture and horticulture
- increasing the oxygen level of ponds used for animal drinking
- stirring biodynamic preparations and soil biology improving agents

Technical data

- Flowrate: 200 lit/min
- Length: 95 cm/Vortex unit
- Width: 90 cm/ Vortex unit
- Height: 32cm/ Vortex unit
- Weight: 9 kg/ Vortex unit
- Material: fiber glass reinforced food safe plastic
- Price: USD560/single unit



VORTEX

ONGOING FLOWFORM® RESEARCH

Flowform® not only oxygenates water effectively, with low capex and opex outlay, it also improves water quality in areas that modern science is only recently understanding but which nature has relied on and created through complex dynamic flow patterns for eons.

The restructuring and energizing of water is something vital for any food growing and processing industry, giving rise to improved product quality.

Scientific research in light frequencies in water indicates that Flowform's figure8 dynamic water flow increases energy in water.



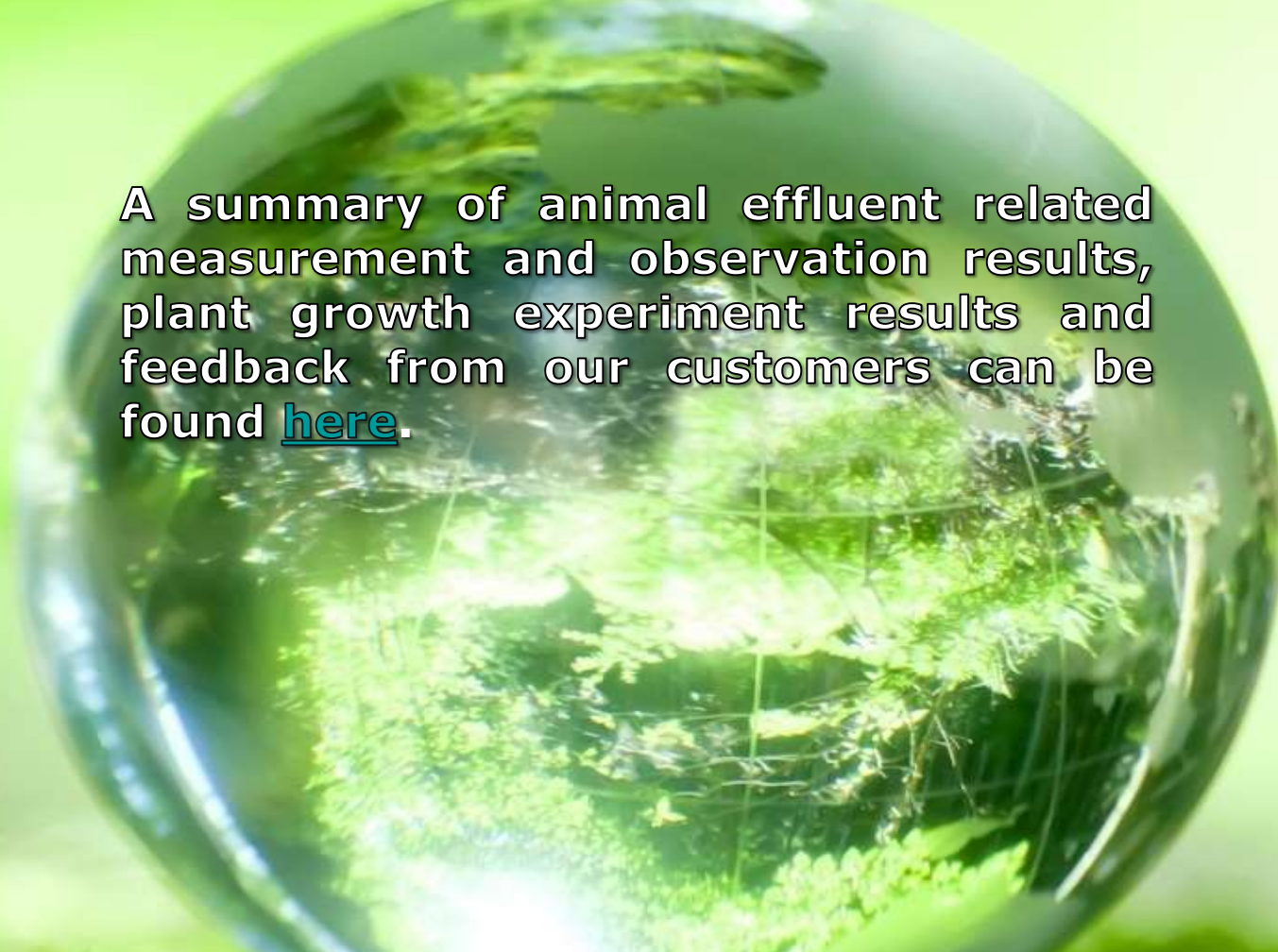
ONGOING FLOWFORM® RESEARCH continued

Increased natural energies in water (called quantum fluctuations) means such water can deliver 'more work', which in nature means increased wellness and a more balanced form

Specific Flowform® research using published infrared imaging (IRI) methods, indicates the method can create highly coherent states in all types of water. (Researchers include Davenas, Ludwig, Schiff, Gross, Del Giudice, Preparata, Pollack and Johansson.)



New Vortex Prototype



A summary of animal effluent related measurement and observation results, plant growth experiment results and feedback from our customers can be found [here](http://www.flowform.net).

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The issue in depth

Lack of oxygen

When high organic content primary effluent is mixed with natural water streams, a natural biological decomposition takes place due to microorganisms living in water. Due to the oxygen consumption of the microorganisms, the water is soon depleted of oxygen leading to the death of inferior and higher organism living in the water. This often results in deterioration of water and the ecosystem nearby.

Anaerobic behavior

Due to the decomposing activity of microorganisms living in **anaerobic** organic waste, pungent smelly and toxic gases are generated. The anaerobic microorganisms produced will start the cleaning process of the sludge through putrefaction of the organic matter, the big molecules (protein, carbohydrate, fat) turn into methane, CO₂ and water.

Aerobic behavior

Decomposing process of organic matter under **aerobic** situation (in presence of oxygen) is called fermentation where aerobic microorganisms are decomposing substrates into CO₂ and water. To start this process you require great amount of air or pure oxygen to be ingested into water.

Should you be interested in more details about the process, click [here](#).